

BMJ Open Quality Improving hydration among hospitalised older adults in an acute geriatric ward with a bundled multi-component intervention

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ABSTRACT

Dehydration compromises patient safety. Hospitalised frail older adults are at risk of dehydration, which can result in significant morbidity and mortality. However, current clinical practice workflows do not adequately address the issue of poor oral hydration. This study aims to analyse this problem and test solutions to increase the percentage of hospitalised older adults who consume at least 1 L of oral fluid per day from 15% to 60% in an acute geriatric ward over 3 months. A work group was formed to conduct this service improvement project. Extensive literature was searched and reviewed, brainstorming of workflow gaps along with discussions was carried out, and strategic solutions were developed. A series of implementations was carried out sequentially and cumulatively to target vital root causes. The implementations include the introduction of a bedside hydration flip chart, individualised fluid schedule chart, hydration rounds, posters to create a visual workplace and regular in-service talks for healthcare staff. The number of patients who met the minimum fluid intake was collected daily over 4 months to monitor for intervention success and sustainability. Implementing a bundled multi-component intervention increased the percentage of older adults who meet the minimum oral fluid intake of 1 L/day by 45%, making it a needful part in the management of geriatric inpatients to prevent dehydration.

INTRODUCTION

Problem

Inadequate oral fluid intake among hospitalised older adults is a growing problem worldwide and results in significant morbidity and mortality.^{1–3} With age, the human body undergoes significant physiological changes, such as reduced thirst sensation and kidney function. These changes, coupled with reduced total body water, deplete fluid reserves, thereby increasing the older adult's risk of dehydration.⁴ Clinical dehydration refers to the loss of body water, with or without salt, at a rate greater than the body can replace. In contrast to volume depletion, which refers to the loss of extracellular water

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Dehydration is a prevalent problem among hospitalised geriatric patients which poses a significant patient safety issue and is increasingly becoming a significant cause of mortality and morbidity among older adults. However, there is a lack of strategies to combat dehydration among older adults in the acute inpatient setting.

WHAT THIS STUDY ADDS

⇒ This service improvement project demonstrates that simple strategies can be effective tools to combat dehydration in hospitalised geriatric patients. The introduction of a multi-component intervention involving a bedside hydration flip chart, individualised fluid schedule chart, hydration rounds, posters and regular educational talks to healthcare staff was successful in increasing the percentage of hospitalised geriatric patients who achieved the minimum fluid intake required to reduce their risk of dehydration.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The strength of a multi-component nursing care bundle used in this project lies in its simplicity, cost-effectiveness and easy applicability in clinical settings. These interventions could be integrated into nursing standards of care and generalisable to benefit older adults admitted throughout the hospital by ensuring safer patient care delivery.



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in the interstitial and vascular compartments, dehydration refers to the loss of body water from the intracellular component.⁵ The Hydration and Outcome in Older Patients admitted to hospital study showed that 37% of those admitted to hospital were dehydrated based on serum osmolality and 62% remained dehydrated when reviewed at 48 hours of admission.² Furthermore, it revealed that patients who were dehydrated at admission were six times more likely to

die in the hospital than those with adequate hydration.² Likewise, clinical dehydration is associated with an increased risk of delirium,^{6 7} acute kidney failure, urinary tract infections⁸ and falls,⁹ which delays recovery, prolongs hospitalisation and poses greater mortality risk.^{3 10} Therefore, dehydration in hospitalised older adults reflects poor care.

In an acute geriatric ward within Singapore General Hospital, 80% of the patients are frail older adults with multiple medical conditions, physical (ie, visual and hearing impairment) and cognitive disability, and complex care needs. These individuals are at increased risk of dehydration and would require effective strategies and a robust carer support system for dehydration prevention. Therefore, prevention, early recognition of risk factors and timely treatment of dehydration in older adults impact patient safety and overall quality of care by mitigating complications. However, ensuring adequate hydration of institutionalised older persons is often challenging, with multiple confounding barriers to overcome.¹¹ The project aims to increase the percentage of geriatric inpatients with optimal oral hydration from 15% to 60% over 3 months in an acute geriatric ward—a step towards preventing dehydration, which is in line with our organisational goals to target zero harm and improve patient outcomes and experience.

Background

Previously, a service improvement project in Singapore Changi General Hospital (CGH) was carried out in 2020, trialling the use of a dysphagia cup to encourage oral intake in geriatric inpatients. The study found that 90% of geriatric patients did not consume the recommended daily fluid allowance of 1.6L/day for females and 2.0L/day for males as recommended by ESPEN guidelines.^{4 12 13} With similar problems of inadequate oral hydration among geriatric inpatients in a local institution, the CGH study provided valuable insight into the potential issues that our study could face. The CGH study noted that the mean amount of water consumed daily by hospitalised older adults was 1030 mL (range of 500–2200 mL), with only 10% meeting the lower limit of daily recommended water consumption.¹³ Therefore, when caring for hospitalised older adults, the practical approach to aiming for optimal hydration is to achieve the best oral fluid intake possible under ongoing circumstances even though it may not meet the recommended daily fluid allowance as per guidelines.^{4 14} A practical target when aiming for the best daily oral fluid intake among hospitalised older adults was 1L based on the mean amount of water consumed in our local geriatric inpatient population and literature.^{7 12 13} Hence, the study aimed to investigate the percentage of geriatric inpatients who consume at least 1L of fluids per day, analyse the problem of poor oral hydration, identify workflow gaps, then generate and test solutions that would improve oral hydration.

METHOD

The Standards for Quality Improvement Reporting Excellence 2.0 guidelines¹⁵ were used for reporting this service improvement project. The population for this project includes patients 65 years old and above who can take orally and are excluded from those on fluid restriction or any form of tube feeding.

Baseline measurement

Fluids were either served in variable sizes of measuring cups or a standard cup with a known fixed volume to facilitate the charting of intake more accurately, which is the main outcome measure. Therefore, patients who can self-manage their fluid balance chart are equipped to easily identify and record the volume of fluid intake. Special note was taken of the patient's preferred drinks to be served at a certain time of the day and at an appropriate temperature, and this information was incorporated into their individualised fluid schedule chart. As toileting needs are a crucial component of hydration care, patients are given an orientation of the toilet location nearest to their bed on admission to the ward, on how to call for assistance to get to the toilet or for diaper change using the bedside call button and are reminded at each nursing shift changeover to always place the call button within reach.

As our nurses are directly involved in creating, implementing and evaluating patient care plans together with the doctors, they play a critical role in ensuring patient safety and delivering quality care. A pre-project survey was conducted to verify nurses' knowledge about dehydration. Out of 78 nurses who participated, 98% understood that frail older adults require timely and adequate hydration to maintain electrolyte balance and well-being. Initial data collections on the oral fluid intake of patients within an acute geriatric ward during a work week period (9 days) provided a 'snapshot' of the problem. This data showed that only a mean proportion of less than 15% met the minimum daily target of 1L fluid consumption over a period of 9 days. Despite having good awareness and adequate knowledge about dehydration in older adults among healthcare providers, poor oral intake among hospitalised older adults persists and is inadequately addressed with current clinical practice methods. Therefore, a pilot quality improvement project was initiated to examine and address root causes for poor fluid intake in this population.

Design

The hydration work group was formed, comprising a geriatrician (co-leader), a gerontology-trained senior nurse clinician (leader) and three staff nurses, and under the guidance of a quality improvement coach. This work group met weekly face-to-face. Other stakeholders were included either in the meetings or interviewed by work group members. Possible root causes for poor oral fluid intake in this population were identified using a cause-and-effect analysis. Root causes were verified through on-site

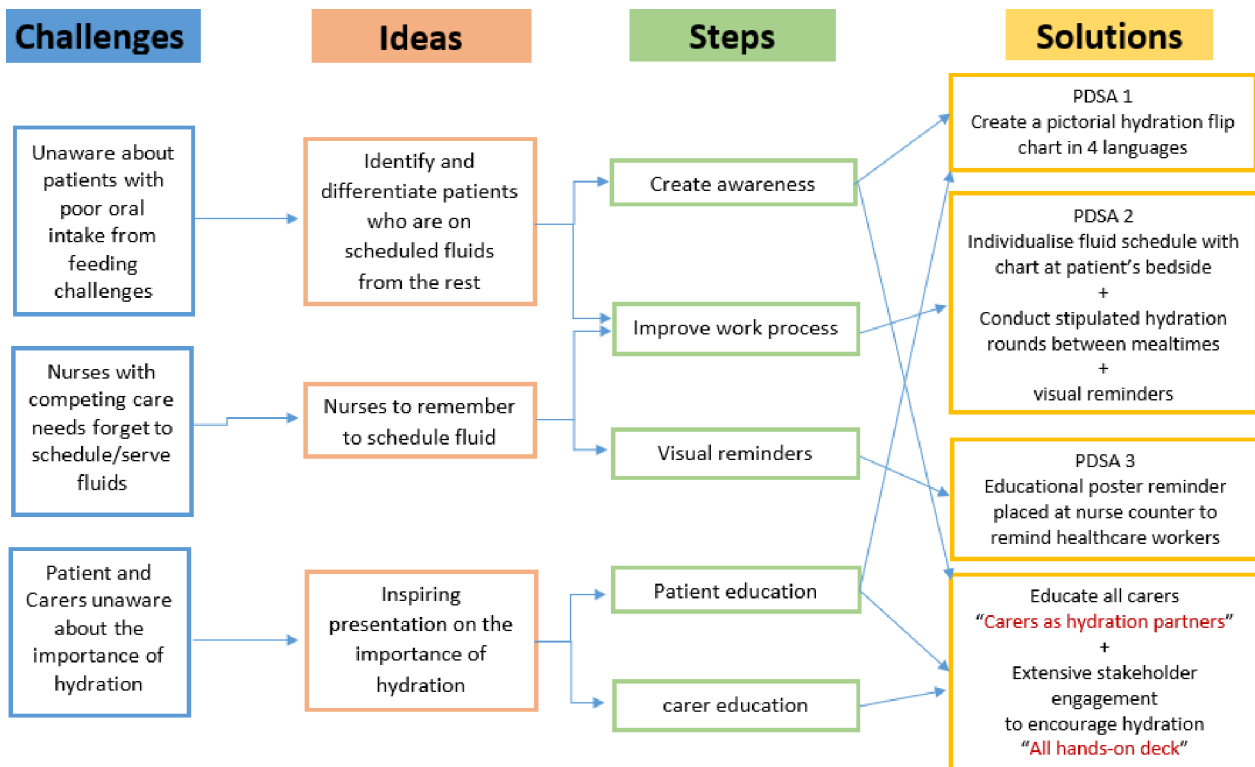


Figure 1 An extended driver diagram (includes final solutions obtained via the prioritisation matrix). PDSA, Plan-Do-Study-Act.

observations and discussions with ward nurses, patients and caregivers. The Pareto Chart of reasons for poor oral intake is a cumulative line graph of all the votes by the work group members and is used to select the top causes that received 80% of the votes. Therefore, the root causes were distilled to three vital ones to focus efforts based on the final Pareto Chart. Thereafter, ideas generated were prioritised based on the level of impact on the goal and difficulty of the task to focus the discussion and achieve consensus. The driver diagram was used to further develop these ideas and to systematically identify possible solutions for the vital root causes. Potential solutions were discussed with patients, next-of-kin, caregivers, ward nurses, geriatricians and dietitians, and feedback was obtained. In [figure 1](#), a simplified version of the driver diagram was presented to only include the shortlisted solutions to be piloted after having generated them from the change ideas. These solutions were selected using a prioritisation matrix based on their effectiveness, ease of implementation and sustainability. A workflow protocol was developed with the involvement of the key stakeholders. As per protocol, once the geriatrician has highlighted a patient to have poor oral hydration and ordered scheduled fluids, the nursing staff will put up the hydration flip chart, the scheduled fluid chart and a nursing handover for closer monitoring of fluid intake. New care providers and nursing staff would receive orientation from nurse and physician champions and would be included in education meetings.

Strategy

The strategy for implementation was based on our driver diagram (in [figure 1](#)). Each change idea was structured

using a Plan-Do-Study-Act (PDSA) test cycle, in which we established the hypothesis to be tested by the change idea and planned the delivery and measurement of the solution. Data were collected continuously throughout the project and the data generated at each test cycle were closely reviewed. The strategy was a planned series of three sequential and cumulative PDSA test cycles (in [figure 2](#)) with ongoing staff engagement via educational talks to target vital root causes concurrently.

Regular educational talks delivered by a consultant geriatrician were conducted for the ward nurses and doctors, covering topics surrounding dehydration in older adults. These topics included age-associated physiological changes and factors that predispose the hospitalised older persons to dehydration, the impact of dehydration on them which affects timely recovery from their acute illness and interventions to prevent dehydration. The impact of dehydration in the hospitalised older adult was also highlighted during weekly multi-disciplinary team meetings, where ideas to identify and improve possible workflow gaps leading to dehydration among patients were further brainstormed.

PDSA cycle 1 (change initiated on 19 November 2021): our initial intervention was to provide a reminder for patients to hydrate and allow healthcare providers to visually differentiate patients on scheduled fluids from others during hydration rounds. All stakeholders involved were agreeable, hence this was done by placing a bedside hydration flip chart with a simple picture and text based on the patient's language literacy (including English, Malay, Mandarin and Tamil), which indicates 'Hi, I need

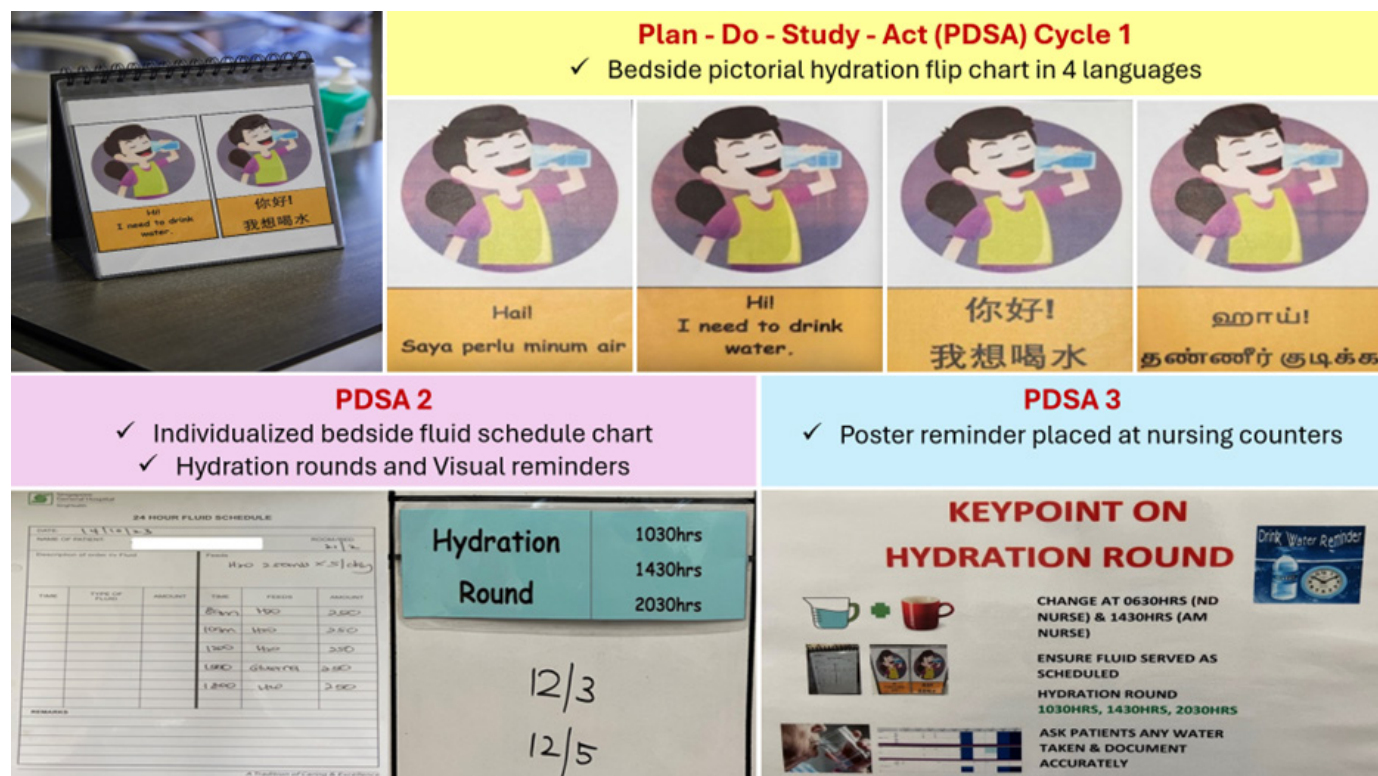


Figure 2 Strategy: a stepwise multi-component bundled intervention.

to drink water' as illustrated in [figure 2](#). This achieved an increase in the percentage of those with optimal hydration to 25%. Feedback received indicated that this was a simple yet sustainable change, and concerns on infection control with the recycling of hydration flip charts were addressed. Hence, the intervention was continued.

PDSA cycle 2 (change initiated on 3 December 2021): to build on the reminder idea, an individualised fluid schedule chart was added to the bedside flip chart, the hydration round schedule was placed on the nursing counter board and incorporated routine hydration rounds in between regular mealtimes for our second intervention. The fluid schedule chart specified the volume of water or other non-caffeinated beverages scheduled to be taken at a particular time, while the hydration rounds provided added opportunity for nurses to coax patients to drink and adhere to the hydration regime, and to document the intake accurately. There was a slight improvement with this, bringing up the percentage of those with optimal hydration to 46%. Ward doctors' feedback was that patients were better hydrated, and nurses felt it made nursing care more systematic and convenient, though there was a slight notable increase in workload, it was still manageable and hence intervention was kept.

PDSA cycle 3 (change initiated on 17 December 2021): we hypothesised that our change idea may be more effective if we visually share information about our ward operations to make human performance safer, more accurate, reproducible, and reliable and efficient. Therefore, we designed hydration round posters (as depicted in [figure 2](#)) and placed them at nursing counters to provide

a visual workplace for nurses who conduct the hydration rounds. The poster was simply worded and accompanied by pictures to emphasise key messages during hydration rounds, such as the timings to change water jug and for hydration rounds, and to serve fluids as scheduled. The nurses and caregivers found it very beneficial, especially amid competing demands and the risk of overlooking the scheduled rounds, and this was also continued.

RESULTS

In 2021, a service improvement study was conducted over a period of 4 months in an acute geriatric ward on patients aged 65 years and above who can take orally, and 233 patients were included. These patients were noted to have poor oral fluid intake on admission by the geriatric medicine team and were identified to be at increased risk of dehydration, while meeting the rest of the inclusion criteria for the study. Our main outcome measure was the percentage of older adults who achieve optimal hydration as defined by 1 L/day. Data collection for this project commenced from 5 November 2021 till 4 March 2022 and the progress following each intervention was presented as a run chart in [figure 3](#). Each intervention cycle has proven to be beneficial in improving our main outcome, as a shift was seen in the run chart after implementing the bundled multi-component intervention.

After PDSA cycle 1, the percentage of geriatric patients who met their daily fluid intake of at least 1 L/day increased by 10%, from 15% to 25%. Adding on PDSA cycle 2, this percentage increased by 21%, from 25% to

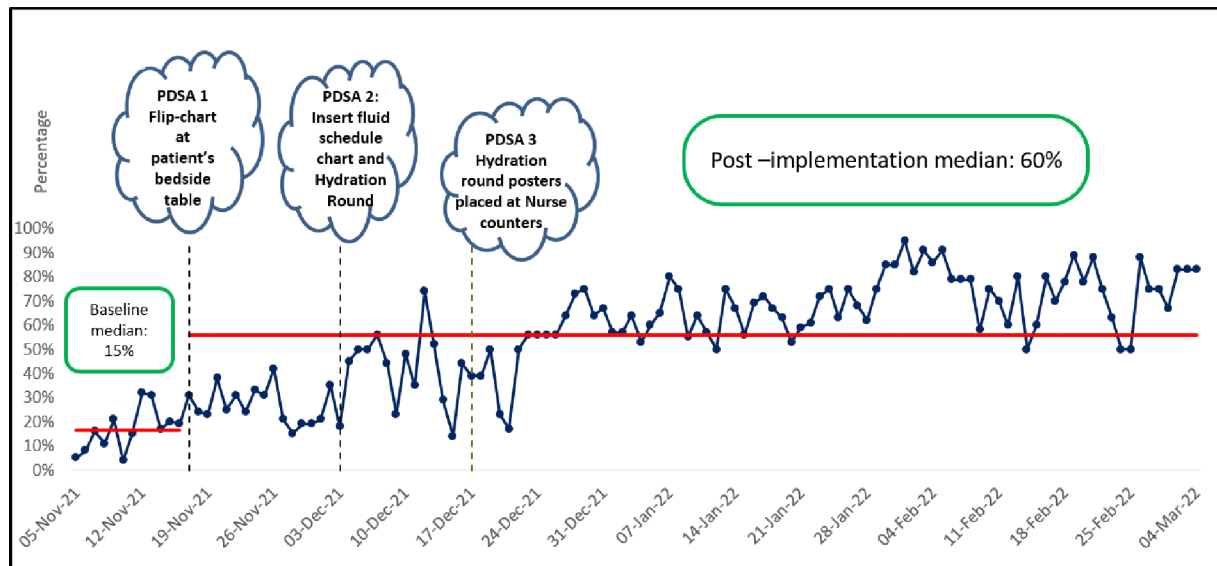


Figure 3 Percentage of hospitalised older adults meeting optimal fluid intake over 4 months. PDSA, Plan-Do-Study-Act.

46%. In addition, patients provided feedback that the flip chart encouraged them to participate in their own care management, which provided them with confidence and a sense of autonomy. Seeing the fluid schedule chart allowed caregivers to appreciate the importance of having adequate fluid intake to help in the patient's recovery. Nurses also expressed that having a physical fluid chart allowed them to easily access a patient's fluid schedule without having to refer to the doctor's note on the computer. This provides added convenience for nurses to serve non-caffeinated beverages at the patient's bedside.

After PDSA cycle 3, the percentage of geriatric patients who had achieved target fluid intake per day increased by 14%, from 46% to 60%. Furthermore, the medical team also witnessed an increase in fluid consumption among geriatric patients post implementation of the bundled multi-component intervention, which had stabilised electrolyte abnormalities noted on admission and the patient's overall condition. To maintain study fidelity, internal audits of workflow, including accurate and timely fluid charting, documentation and handovers, were conducted regularly by the nurse and clinician champions of the project.

Interestingly, post implementation of the above strategy, there was a notable decrease in restraint use in the acute geriatric ward from 7.5% to 3.1%. This is in keeping with what is known, that adequately hydrated patients are less likely to present with behavioural episodes,^{9 12} hence a lesser indication for restraint use. Without being restrained, patients' well-being is maintained and they are more likely to perform basic activities of living independently and participate in rehabilitation therapy sessions. Furthermore, the incidence of hospital-acquired pressure injury among this population decreased from 9% to 3.1% post-strategy implementation. With better hydration status, patients have a better chance of recovering from their acute illness without sustaining complications

of prolonged bed rest (ie, pressure injuries and functional decline).^{7 16} Adequate hydration also promotes and preserves the integrity of the skin barrier.⁷

LEARNINGS

Using multi-component bundled interventions, the service improvement project team was successful in raising the percentage of geriatric inpatients who achieved their target fluid intake of 1 L per day from 15% to 60% over 4 months.

The desire to deliver patient-centric care among health-care professionals for better patients' experience, care quality and overall health outcomes led to the development of a service improvement project to improve hydration status among hospitalised older adults. This project highlights the utility of multi-component nursing care interventions to address the multi-dimensional problem of dehydration among hospitalised geriatric patients. Bunn *et al* reported a positive effect on dehydration among geriatric patients with multi-component interventions.¹⁷ By targeting root causes involving multiple key stakeholders, the interventions implemented in this study were naturally more likely to achieve positive results. Simple interventions that were implemented as part of the patient's daily care activities enabled healthcare staff to follow through without compromising the patient's care. Multi-component solutions have been commonly viewed to be resource-intensive and more complex to implement and sustain.^{18 19} However, this study demonstrated that resource-intensive interventions, if made part of the clinical care routines, coupled with all care partners' involvement, would become feasible and sustainable. Moreover, the interventions used in this study were inexpensive to design and readily available for implementation (ie, flip charts and posters were affordable to design and print). The bundled multi-component intervention provides

adequate patient education and facilitates fluid round care activity at the stipulated time schedule.

In addition, education is key in countering dehydration among geriatric patients. Educating patients and their caregivers on the importance of sufficient hydration to aid in their recovery bridged the knowledge gap that they had. This includes refuting the idea that drinking water increases the need to access the bathroom more frequently. Introducing the fluid schedule chart to patients and caregivers empowered them to self-manage. Patients and caregivers' education, coupled with simple solutions, facilitates patients' compliance with the fluid schedule. Equally vital was helping healthcare staff understand the clinical complications and implications of dehydration on patients. With a deeper appreciation of the effects that dehydration could have on patients, healthcare staff could better intervene and educate their patients on the importance of hydration. This is in keeping with the *Behaviour Change Wheel* proposed by Michie *et al*, who highlighted education as one important tool to effect behavioural change.²⁰ It is also worthwhile to note that among the interventions used to improve hydration in the acutely unwell older adult, few addressed the knowledge gap, as reported by Bruno *et al*.²¹

LEARNINGS AND CHALLENGES

This study applies a stepwise approach in implementing interventions, while sustaining continued engagement of stakeholders involved, including healthcare staff, patients and their caregivers. The stepwise approach allowed time for nurses and caregivers to familiarise themselves with each intervention before the next phase was implemented. Furthermore, this teamwork approach towards improving care of the older person was well received by the healthcare staff, patients and their families who were part of the patient's recovery journey. Patient's next of kin proactively reminded and served patient fluids at the scheduled times to ensure that their loved ones adhere to the hydration schedule. Feedback and suggestions were frequently received from nurses and caregivers, which in turn helped to improve and refine the interventions.

In addition, this project has successfully implemented simple, pragmatic and cost-effective solutions. The interventions leveraged existing resources and workflows without causing significant disruption and inconvenience to the various stakeholders. This, in turn, improved buy-in and compliance with the interventions, allowing for sustainable outcomes. Moreover, easy applicability and cost-effectiveness enhance the likelihood of adoption for widespread use in other wards and healthcare institutions, benefiting patients in these settings.

Nonetheless, there were challenges faced while implementing interventions in this study. Increased workload was a common concern raised by ward nurses, as hydration rounds were an add-on to daily nursing duties. When there was competing demand, some nurses inadvertently

forgot to conduct hydration rounds, affecting the adherence to the hydration schedule.

It was also challenging for older adults to cooperate with the hydration schedule, especially those with cognitive impairment. Therefore, patients are encouraged to take small frequent amounts of fluids at each time. Patient's family and caregivers were approached to assist with feeding their loved ones. However, some patients simply refused to cooperate with the nurses or their caregivers despite their best efforts, making it difficult for the measures to be implemented. Some reasons for fluid refusal include the fluid temperature and the timing of the scheduled fluids, which differed from their personal preference and routine, respectively. Finally, as the flipchart folders were reused for new patients, infection control was of concern. Therefore, the flipcharts required regular disinfection. Challenges were addressed along the way with extensive stakeholder engagement, education and reassurance.

Implications for practice and research

Sustainability is key. The work group continues to innovate, improve on current interventions and spread the change culture. Likewise, to integrate these successful interventions into routine clinical practice, findings were presented at the nursing leaders' monthly meeting, which has garnered interest for wider implementation across other wards in our institution. As the work gained traction, the importance of adequate hydration in older adults and collaborative effort among all carers was highlighted and featured in the local institution newsletter with this tagline 'Carers are hydration partners'. Furthermore, an easy-to-read educational material on 'dehydration prevention' was also developed and uploaded onto patients' bedside iPads, allowing patients and their caregivers to refer to at their convenience and for continued education in the absence of a healthcare provider. These efforts help supplement the interventions discussed in this paper and promote continued adoption of the solutions to ensure sustained effects in the long term. Moving forward, a greater multi-disciplinary involvement would enhance the effectiveness of the interventions, where medical, allied health professionals and nursing staff complement each other by encouraging the hydration of patients. In addition, regular awareness campaigns may complement the existing efforts and future research could focus on analysing the benefits of these interventions with regards to length of stay and readmission rates.

CONCLUSION

Optimal hydration during hospitalisation is an important indicator of healthcare safety and quality provided by any healthcare institution. The bundled multi-component intervention increased the percentage of older adults who met the best daily oral hydration by 45%. Therefore, these interventions, when adapted as standard of care practice, would ensure safer patient care delivery.

Nonetheless, preventing dehydration in this population requires a multi-disciplinary team effort. Healthcare providers and caregivers have a vital role in educating patients on the importance of hydration to improve their knowledge and health literacy, recognising early warning signs and risk factors of dehydration in frail older adults, and optimising care. Furthermore, practical approaches are necessary to motivate older adults to drink enough, such as encouraging frequent rather than large quantities of hydration, and serving a variety of non-caffeinated beverages instead of one.^{22 23} Therefore, an integrated patient-centric approach is crucial for delivering quality inpatient care.

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Contributors AM, RQMN, WFC, SGES, NA conceptualised and designed the study, developed methodology, performed formal analysis, investigation, created visualisations and managed project administration. SLLT and RQMN was involved in literature search and preparing the first draft of the manuscript. RQMN, LX and MKCN were involved in providing expert advice and input for the project. AM, WFC, SGES, NA were involved in data collection, educating the nurses, and setting up workflow. All authors have contributed to the design of the work, reviewed, and edited the manuscript, and approved the final version of the manuscript. AM and RQMN provided overall supervision and served as guarantors of the study.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This is a service improvement project without research intent (research outcomes) and not posing additional risks or burden to the subjects. Therefore, this project did not require Institutional Review Board (IRB) review and approval prior to its conduct. The project was conducted in an ethical manner and compliant to all applicable policies, acts (such as PDPA) and regulations within the institution where it was carried out. Patient anonymity was preserved.

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